

WebLens

Website Performance & SEO Intelligence Platform

Project Overview

WebLens is a web-based website analysis platform that helps developers, businesses, and website owners evaluate the quality of their websites through automated audits. The platform analyzes website performance, search engine optimization (SEO), accessibility, security, and best practices, then provides actionable recommendations for improvement.

WebLens transforms complex website metrics into easy-to-understand insights, enabling users to optimize website speed, user experience, and search engine visibility.

Problem Statement

Many website owners are unaware of issues affecting their website's performance, SEO ranking, accessibility compliance, and security. Existing auditing tools often present technical data that is difficult for non-technical users to understand.

WebLens addresses this problem by generating comprehensive website audits and providing clear recommendations that help users improve their websites.

Objectives

- Analyze website performance
 - Evaluate SEO readiness
 - Detect accessibility issues
 - Identify security vulnerabilities
 - Generate actionable optimization recommendations
 - Provide downloadable audit reports
 - Track website improvements over time
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Target Users

Developers

- Analyze website performance
- Improve Core Web Vitals
- Optimize frontend resources

Freelancers

- Audit client websites
- Generate professional reports
- Demonstrate optimization opportunities

Businesses

- Monitor website health
- Improve user experience
- Increase search visibility

Students

- Learn website optimization techniques
 - Understand web performance metrics
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Core Features

1. Website URL Analysis

Users enter a website URL and initiate a complete website audit.

Example:

`https://example.com`

System validates URL before processing.

2. Performance Analysis

Evaluates website loading speed and user experience metrics.

Metrics analyzed:

- Performance Score
- First Contentful Paint (FCP)
- Largest Contentful Paint (LCP)
- Speed Index
- Total Blocking Time (TBT)
- Cumulative Layout Shift (CLS)

Output example:

Performance Score: 91

FCP: 1.1s

LCP: 2.3s

CLS: 0.05

TBT: 120ms

3. SEO Analysis

Evaluates website search engine optimization readiness.

Checks include:

- Meta title
- Meta description
- Heading hierarchy
- Sitemap availability
- Robots.txt
- Mobile responsiveness
- Open Graph metadata
- Canonical URLs

Output example:

SEO Score: 94

- ✓ Meta Description Found
 - ✓ Sitemap Available
 - ⚠ Missing Open Graph Image
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4. Accessibility Analysis

Evaluates usability for users with disabilities.

Checks include:

- Alternative image text
- ARIA labels
- Form accessibility
- Color contrast
- Keyboard navigation
- Semantic HTML structure

Output example:

Accessibility Score: 82

- ⚠ 4 Images Missing Alt Text
 - ⚠ Low Contrast Button
 - ✓ Keyboard Navigation Supported
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5. Security Analysis

Evaluates website security configuration.

Checks include:

- HTTPS usage
- SSL certificate validity
- Security headers
- Mixed content detection
- Content Security Policy
- X-Frame-Options
- HSTS configuration

Output example:

Security Score: 88

- ✓ HTTPS Enabled
 - ✓ SSL Certificate Valid
 - ⚠ Missing HSTS Header
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6. Best Practices Analysis

Evaluates adherence to modern web development standards.

Checks include:

- Console errors
- Deprecated APIs
- Resource optimization
- Responsive design
- Modern image formats
- Code efficiency

Output example:

Best Practices Score: 89

- ⚠ Unused CSS Found
 - ⚠ Large JavaScript Bundle
 - ✓ Secure Requests
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7. Smart Recommendations Engine

Provides personalized suggestions based on audit findings.

Example:

Issue:

Largest Contentful Paint exceeds recommended threshold.

Recommendation:

Convert hero images to WebP format and enable lazy loading.

Expected Improvement:

+15 Performance Score

8. Audit Dashboard

Displays all audit results through a modern analytics dashboard.

Dashboard sections:

- Overall Score
 - Performance Metrics
 - SEO Results
 - Accessibility Results
 - Security Results
 - Recommendations
 - Historical Audits
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9. PDF Report Export

Users can export audit results as professional PDF reports.

Report includes:

- Website information
 - Audit summary
 - Performance metrics
 - Identified issues
 - Improvement recommendations
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10. Audit History

Stores previous website analyses for comparison.

Features:

- View previous reports
- Compare audits
- Track improvements
- Monitor website growth

System Workflow

Step 1

User enters website URL.

Step 2

Frontend sends request to backend API.

Step 3

Backend initiates website audit.

Step 4

Audit engine collects website metrics.

Step 5

Results processed and categorized.

Step 6

Recommendations generated.

Step 7

Results displayed on dashboard.

Step 8

Optional PDF report generation.

Technology Stack

Frontend

- React.js
- Vite
- Tailwind CSS
- Framer Motion
- Recharts

Backend

- Django
- Django REST Framework

Database

- PostgreSQL

APIs

- Google Lighthouse
- Google PageSpeed Insights

Authentication

- JWT Authentication

Deployment

Frontend:

- [Vercel](#)

Backend:

- [Render](#)

Database:

- [Neon](#)
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Future Enhancements

AI-Powered Optimization Suggestions

Automatically generate optimization strategies using AI.

Competitor Comparison

Compare website performance against competitors.

Scheduled Audits

Automatically run recurring website analyses.

Team Collaboration

Allow teams to share reports and monitor projects.

Browser Extension

Analyze the currently opened website directly from the browser toolbar.

Expected Outcomes

- Faster websites
 - Improved SEO performance
 - Better accessibility compliance
 - Enhanced security posture
 - Improved user experience
 - Actionable optimization insights
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Project Summary

WebLens is a full-stack website intelligence platform that analyzes website performance, SEO, accessibility, security, and development best practices. It provides clear visual reports and optimization recommendations, helping users improve website quality and user experience through data-driven insights.